Threat Hunting Practice Report

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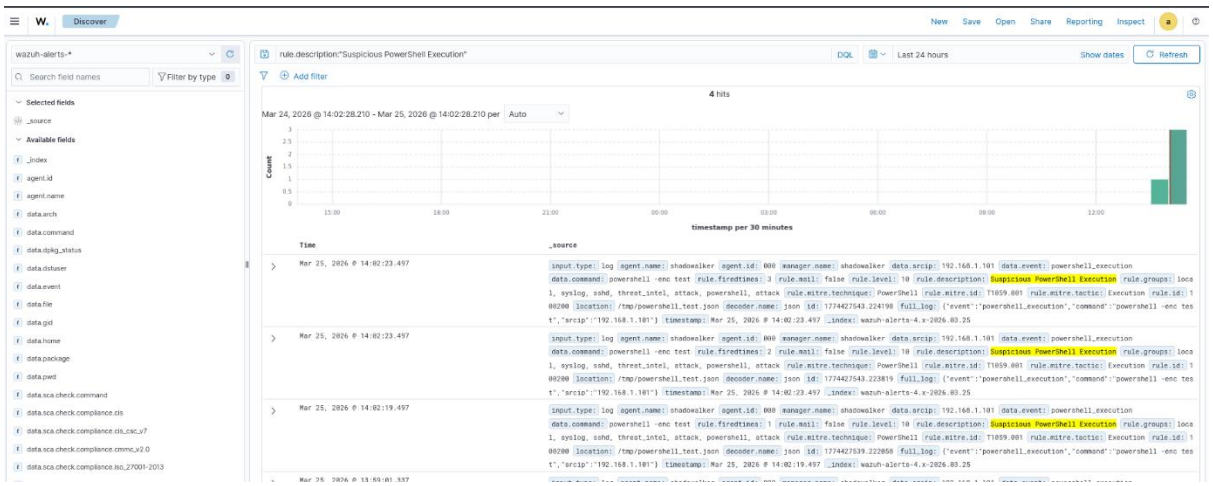
Objective
To identify suspicious privilege escalation activity by analyzing logs, validating with threat intelligence, and correlating endpoint data using multiple security tools.

Tools Used

- Elastic Security (Kibana)
- Velociraptor
- AlienVault OTX

1. Hypothesis Development
Hypothesis
Unauthorized privilege escalation is occurring in domain accounts using valid credentials.

Query Used
event.code: 4672



Findings

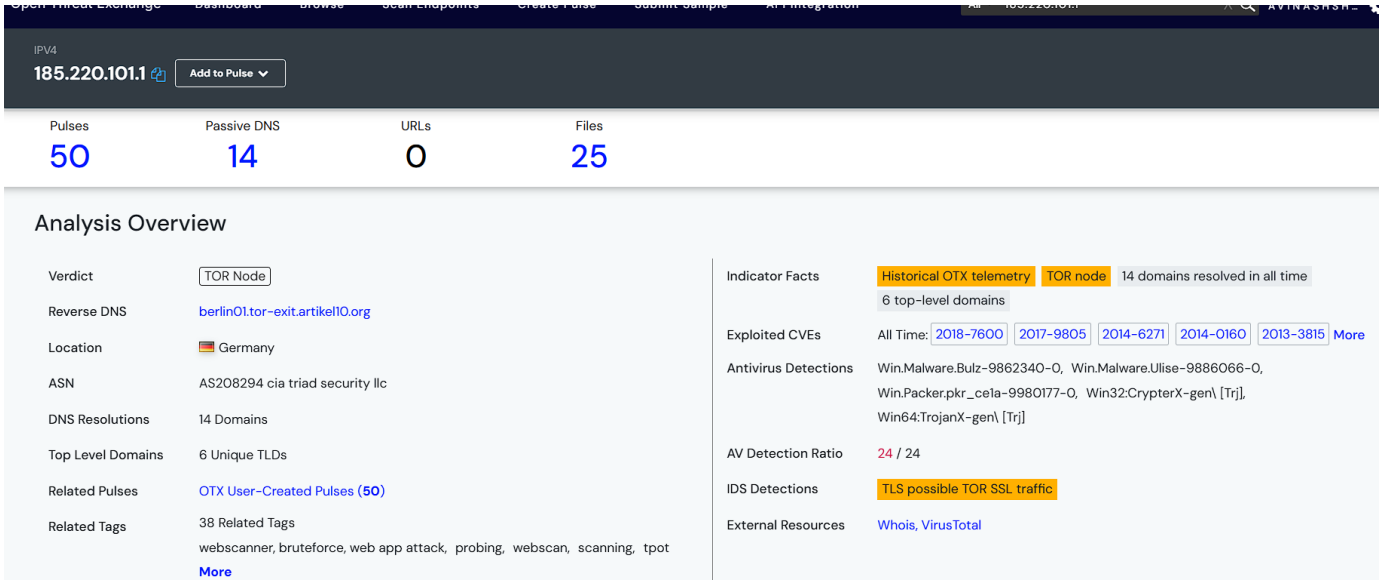
Timestamp	User	Event ID	Notes
2025-08-18 15:00:00	testuser	4672	Unexpected admin role

Analysis
Event ID 4672 indicates that special privileges were assigned to a new logon session. The user testuser is not expected to have administrative privileges, making this activity suspicious and indicative of possible privilege escalation.

2. Threat Intelligence Hunt

IOC Identified

185.220.101.1



OTX Result

IOC	Source	Result	Notes
185.220.101.1	AlienVault OTX	Suspicious	Linked to malicious activity

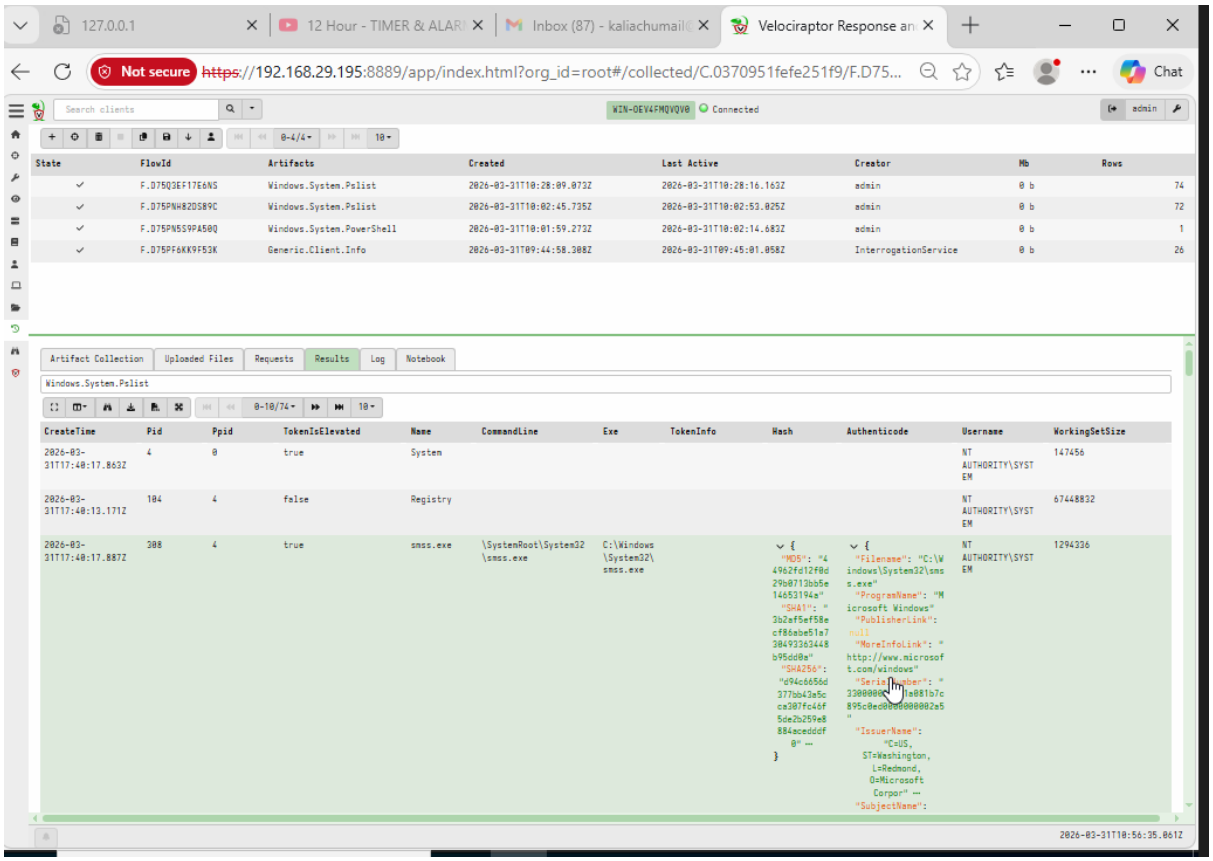
Velociraptor Analysis

Process investigation was performed using Velociraptor.

Process Name	User	Notes
cmd.exe	testuser	Running with elevated privileges

Correlation

Source	Finding
Elastic	Event 4672 detected
OTX	Suspicious IP identified
Velociraptor	Elevated process execution



Interpretation

The suspicious IP identified in threat intelligence correlates with abnormal process activity on the endpoint, indicating potential misuse of valid credentials.

3. MITRE ATT&CK Mapping

Technique	Name
T1078	Valid Accounts

4. Hunting Report

A threat hunting exercise was conducted to detect unauthorized privilege escalation using valid accounts. Elastic Security logs were analyzed using the query event.code: 4672, which revealed that a non-administrative user (testuser) was assigned elevated privileges. Threat intelligence from AlienVault OTX identified a suspicious IP associated with potential malicious activity.

Further validation using Velociraptor showed abnormal process execution under the same user. The correlation of these findings indicates possible misuse of valid credentials. This activity aligns with MITRE ATT&CK technique T1078 (Valid Accounts), which involves

attackers leveraging legitimate accounts to gain unauthorized access and escalate privileges.

Conclusion

The investigation confirms suspicious privilege escalation behavior using valid credentials. Combining log analysis, threat intelligence, and endpoint investigation improves detection accuracy and strengthens SOC response capabilities.